

Automated lumbar intervertebral disc identification and herniation detection in MR images using cascade CNN architecture

Md Abu Sayed^a, Ashiqur Rahman^a, Sadman Mohammad Nasif^b, Sudipto Halder^c, Akram Hossain^a, Hasan Ahmed^d, Muhammad Abdul Kadir^{a,*}

^a Department of Biomedical Physics and Technology, University of Dhaka, Dhaka, 1000, Bangladesh

^b Department of Computer Science Engineering, Khulna University of Engineering & Technology, Khulna, 9203, Bangladesh

^c Department of Mechatronics Engineering, Khulna University of Engineering & Technology, Khulna, 9203, Bangladesh

^d Technical Technical Expert-Environment, UNOPS South Asia Multi-Country Office, Dhaka, 1212, Bangladesh

ARTICLE INFO

Keywords:

Disc herniation
Lumbar intervertebral disc
IVD
YOLOv11
ROI detection
Attention module
VGG19
Cascade CNN

ABSTRACT

Objective: Identifying herniated discs in MRI scans is inherently challenging due to the small size, irregular shape, and complex appearance of the affected regions. Conventional approaches typically rely on semi-automated region-of-interest (ROI) selection and single-model classification using either axial or sagittal views, limiting diagnostic performance. This study aims to develop an automated, accurate, and efficient system for the detection and classification of lumbar intervertebral disc herniation using deep learning models applied to axial and sagittal MR images.

Methods: A YOLO-based framework was developed to automatically identify lumbar intervertebral discs (IVD₁₋₅) and extract ROIs from MR images. Attention-enhanced and fine-tuned VGG19 and ResNet50 models were employed to analyze axial and sagittal images for herniation detection. A decision fusion strategy was used to combine the classification probabilities from both models to further enhance accuracy. The dataset underwent extensive preprocessing and augmentation to improve model robustness and generalization.

Results: The proposed approach demonstrated exceptional performance in detection and classification tasks. For detection, the model achieved mAP50 scores of 95.18 % (axial IVD₁₋₅), 99.50 % (lumbar regions), and 94.87 % (sagittal IVD₁₋₅). Classification accuracy reached 97.05 % for axial images and 97.45 % for sagittal images, increasing to 98.09 % with decision fusion.

Conclusion: Designed to assist physicians, especially during high-demand periods such as pandemics, this approach has the potential to improve diagnostic efficiency and reduce clinical workload.

1. Introduction

Lower back pain (LBP) is a global health concern, affecting millions of people across continents regardless of **geographical** boundaries or socioeconomic **status**. According to the World Health Organization (WHO), approximately 619 million individuals worldwide suffer from some degree of lower back pain [1]. Among the various causes contributing to LBP, disc herniation **stands out** as a significant etiological factor, **accounting** for a substantial proportion of cases [2]. LBP due to disc herniation is a prevalent musculoskeletal disorder in the world, with millions affected daily. Its impact is far-reaching, beginning with healthcare workers struggling to care for LBP patients and extending to thousands of hours of productivity loss in the financial

calculations of the global economy. Particularly vulnerable are individuals living below the poverty line, whose livelihoods often depend on physically demanding labor [3]. For these populations, disc herniation can be especially debilitating. Thus, timely and accurate diagnosis of disc herniation is essential to ensure proper treatment. Disc herniation, also referred to as a prolapsed lumbar intervertebral disc (PLID), is a medical condition in which the nucleus pulposus (the soft inner core of an intervertebral disc) protrudes through a weakened or ruptured annulus fibrosus. This results in compression of adjacent neural structures, consequently leading to pain, sensory disturbances, and motor deficits [4]. For the diagnosis of disc herniation, magnetic resonance imaging (MRI) is generally used as the imaging modality [5]. However, current diagnostic practices involve manual interpretation of MR images

* Corresponding author.

E-mail address: kadir@du.ac.bd (M.A. Kadir).

<https://doi.org/10.1016/j.imu.2025.101648>

Received 21 February 2025; Received in revised form 1 May 2025; Accepted 2 May 2025

Available online 3 May 2025

2352-9148/© 2025 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

by radiologists. This leads to several inherent limitations, including subjectivity, inter-observer variability, and inefficiency [6]. To address these limitations, there is a growing interest in automating disc herniation diagnosis using advanced computational methods, particularly deep learning (DL). While individual deep learning models have shown promise in classification tasks, ensemble learning approaches offer the potential to enhance diagnostic accuracy, robustness, and generalizability [7]. Moreover, although convolutional neural networks (CNNs) are effective at extracting important image features, attention-based models further improve performance by focusing on critical regions within the image [8,9].

Over the past decade, substantial progress has been made in applying machine learning (ML) and deep learning (DL) techniques to detect and classify medical images related to lumbar disc herniation. These approaches have demonstrated significant improvements in accuracy and can be broadly categorized into semi-supervised and fully automated methods. In semi-supervised methods, Ebrahimzadeh et al. proposed an automated approach for diagnosing lumbar disc herniation using T2W sagittal and myelographic MR images [10]. The detection process involved Otsu thresholding, followed by a third-order polynomial alignment and region of interest (ROI) detection. Data extracted from the MR images were subsequently used for feature-based classification. Multilayer Perceptron (MLP), Support Vector Machine (SVM), and K-Nearest Neighbor (KNN) algorithms were evaluated on a dataset comprising 30 clinical cases with 210 lumbar discs, achieving classification accuracies of 95.23 %, 92.38 %, and 91.90 %, respectively. Unal et al. developed hybrid models for detecting lumbar disc abnormalities using T2-weighted sagittal and axial MR images from 55 patients [11]. They extracted 27 features and used FSFS and CBFS methods to reduce the feature count to 5 and 22, respectively, for classification via MLP, SVM, Naïve Bayes, Decision Tree, and KNN algorithms. Hou et al. proposed a semi-supervised approach involving random self-supervised pretraining on unlabeled MR images, followed by classification using an MLP [12]. Tested on a public MRI dataset, the model achieved an accuracy of 87.11 %, a specificity of 86.72 %, a sensitivity of 87.50 %, and an AUC of 94.87 %, with heatmaps visualizing degenerative changes. Ghosh et al. introduced a probabilistic model to detect abnormal discs from T2W sagittal MR scans, reaching an accuracy of 94.85 % on 35 clinical cases comprising 175 lumbar discs [13].

In fully automated approaches, Mbarki et al. developed a deep CNN based on VGG16, achieving 94 % accuracy using a U-Net based ROI method on a private dataset [14]. Al-Kubaisi et al. conducted a comparative study of various transfer learning models (with/without fine-tuning), including VGG19, ResNet50, MobileNetV2, and VGG16 [15]. By applying a block-based ROI method, they enhanced the performance of these models, with VGG16 achieving accuracies between 87 % and 90.91 %, and VGG19 achieving accuracies between 85.05 % and 89.91 % on axial MRI data from 181 patients. Similarly, Q. Pan et al. developed a ResNet101 model for three-class classification (normal discs, disc bulges, and disc herniation) for each lumbar intervertebral disc (IVD) from axial MR images and employed Faster R-CNN for ROI detection [16]. Alsmirat et al. focused on identifying the type of disc herniation using a CNN model, achieving an accuracy of 95.65 % with ROI images from axial MR data of 164 subjects. The Scale-Invariant Feature Transform (SIFT) algorithm was used for ROI detection [17]. M.A. Sayed et al. proposed a YOLOv8 model for IVD detection from sagittal MR images, achieving high detection accuracy. They also examined the impact of fine-tuning and ROI-based classification and suggested a weighted ensemble approach for sagittal image classification and segmentation [18].

Despite these advancements, many existing methods face limitations such as dependence on manual feature extraction, inadequate ROI isolation, and the isolated use of axial or sagittal images. These limitations can lead to inaccurate classifications, and many studies have underutilized attention modules and decision fusion strategies for combining insights from multiple image views.

To address these limitations, this study proposes a robust, fully automated method for detecting and extracting each of the five intervertebral discs in the lumbar region (IVD₁₋₅) from both axial and sagittal MR images. The system uses YOLOv11 for IVD detection and introduces attention-enhanced, fine-tuned VGG19 and ResNet50 models for classifying disc herniation from axial and sagittal views, respectively. By combining detection and classification components, the system can identify which specific disc is herniated. Additionally, fusing classification decisions from both models significantly enhances overall diagnostic accuracy. Techniques such as data augmentation, dropout, early stopping, and transfer learning are incorporated to prevent overfitting and improve generalization.

2. Materials and methods

2.1. Collection and labelling of dataset

The MR image dataset was obtained upon request from the corresponding author of the paper [18]. The dataset comprised 1098 mid-axial lumbar IVD MR images, including 650 labeled as No-PLID and 448 as PLID. These images were collected from 220 patients, of whom 210 were diagnosed with disc herniation, while 10 had no significant spinal issues. First, the DICOM images were converted to PNG format using MicroDICOM. Next, the mid-axial MR images of the lumbar spine were labeled as either PLID or No-PLID, based on corresponding sagittal MRI views using the RadiAnt DICOM Viewer and diagnostic reports. The mid-sagittal MR images were also separated into another folder. Multiple lumbar spine MRI series were acquired for each patient, including T2, axial T2*, and T1 sequences. Each series contained multiple slices; however, only the central slice from the axial and sagittal T2 and axial T2* sequences was utilized in the dataset. The annotation was performed using the Roboflow web application to create labels for the YOLO model [19]. The dataset was divided into training, validation, and testing sets in a 75:15:10 ratio. The medical records and MRI images were fully de-identified and anonymized.

Fig. 1 shows the overall workflow for the automated lumbar intervertebral disc identification and herniation detection in MR images. First, the lumbar region was detected and cropped, followed by the identification and cropping of IVD₁₋₅ from sagittal and axial MR images. Finally, the axial and sagittal images were classified as either PLID or No-PLID.

PLID_a = Probability value of PLID predicted by the VGG19 model from axial MR images.

PLID_s = Probability value of PLID predicted by the ResNet50 model from sagittal MR images.

PLID_p = Final classification as PLID for a patient, determined by combining decisions from the VGG19 and ResNet50 models using both axial and sagittal MR images.

If both models (VGG19 and ResNet50) agree on either PLID or No-PLID, the final classification for the patient is based on their consensus. Otherwise, the decision is determined by the model that assigns a higher probability to its classification choice.

2.2. Data augmentation parameters

Applying random positional and color augmentations increased the number of images in the dataset, enhancing its diversity while preserving the core information. These augmentations help prevent overfitting and improve the model's ability to generalize during training. As a result, the dataset size was effectively tripled, further strengthening model performance.

- **Random Vertical Flip:** The images were flipped vertically at random with a specified probability. For any additional regions in the axial images, the "nearest" interpolation mode was applied.

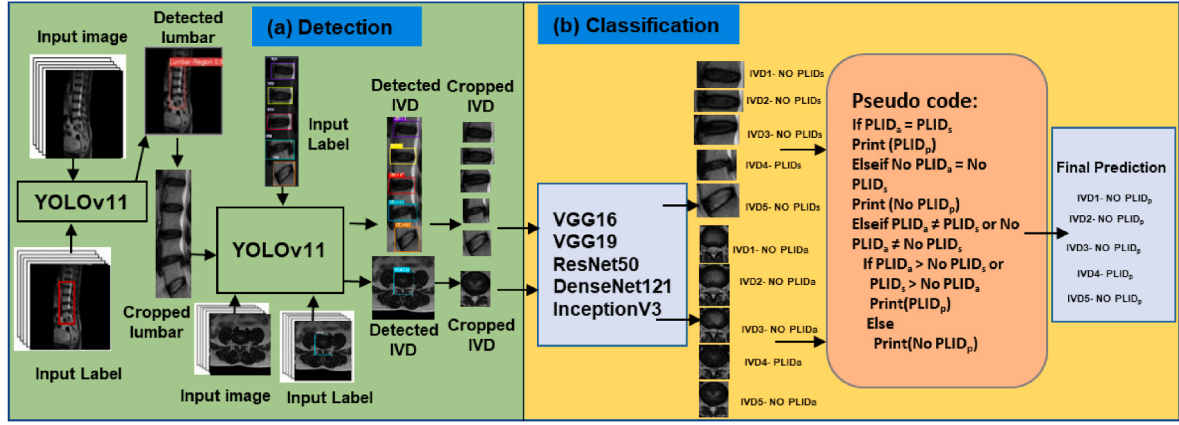


Fig. 1. Workflow for automated lumbar intervertebral disc identification and herniation detection in MR images.

- **Random Rotation:** The images were rotated within a specified range of [0, 30] degrees. The "nearest" mode was utilized to interpolate any additional regions in the axial images.
- **Brightness Adjustment:** The images were either lightened or darkened to introduce variability in brightness. The brightness range applied was [0.1, 0.5].

2.3. ROI detection and cropping

YOLOv11 is a state-of-the-art object detection model designed for high accuracy and efficiency, incorporating several architectural improvements [20]. Its backbone extracts multi-scale features using the C3k2 block and Spatial Pyramid Pooling Fast (SPPF), enhancing object representation across different sizes, as illustrated in Fig. 2. The neck

integrates the Cross-Stage Partial with Spatial Attention (C2PSA) mechanism, refining feature fusion while emphasizing key regions to improve detection accuracy without increasing computational costs. The head utilizes advanced multi-scale prediction techniques for precise bounding box regression, classification, and segmentation. Unlike two-stage detectors such as Faster R-CNN, which first generate region proposals and then classify them, YOLOv11 directly predicts object classes in a single pass. This approach enhances inference speed without compromising accuracy. In the context of lumbar spine MRIs, where multiple intervertebral discs (IVD₁₋₅) need to be identified in a single image, a YOLO-based architecture was selected as it processes the entire image in a single forward pass, making it highly efficient for clinical applications. Intervertebral discs are small and irregularly shaped; their appearance can vary due to orientation, degeneration, or pathology.

Backbone

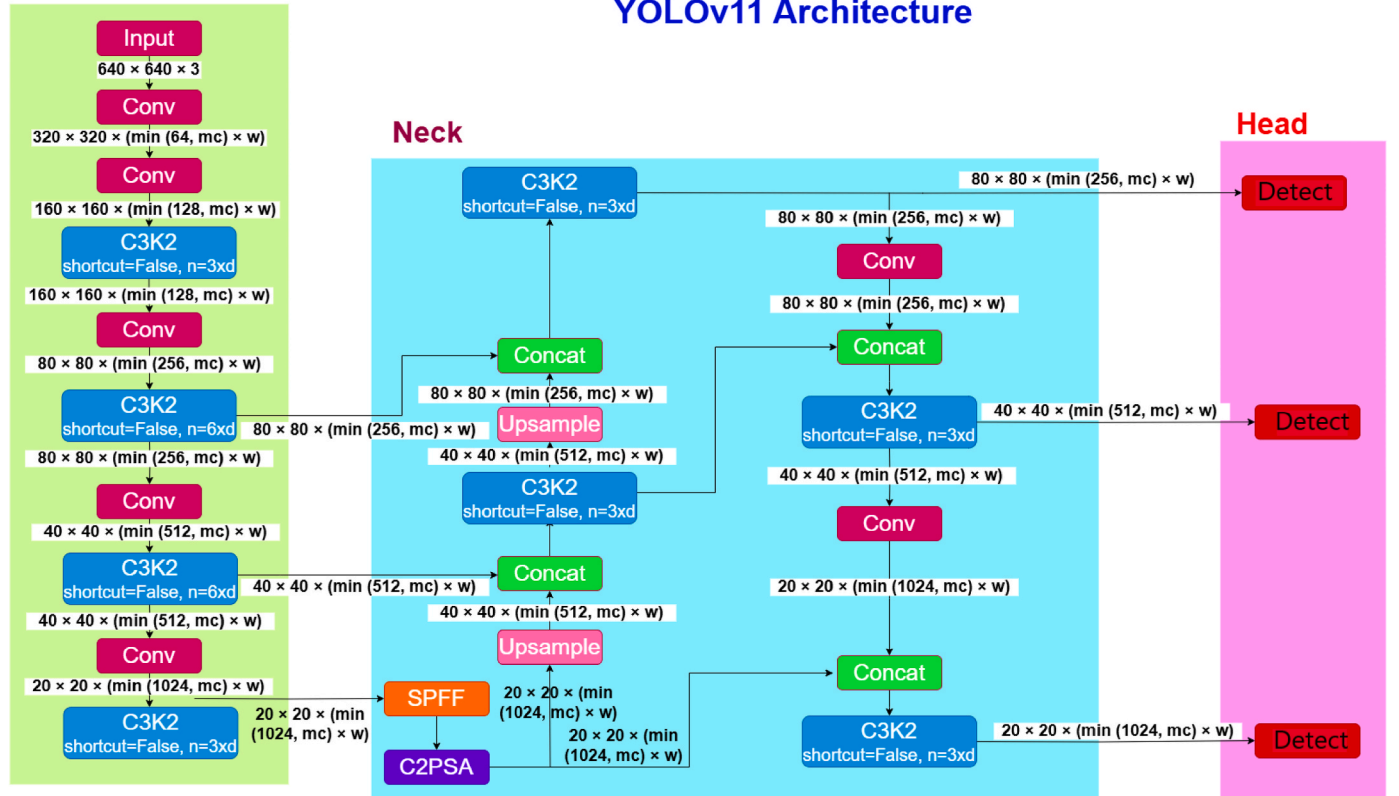


Fig. 2. The YOLOv11 architecture for detecting lumbar regions and IVD₁₋₅ and cropping ROI; reproduced from Ref. [21].

YOLO's ability to detect multiple bounding boxes with high spatial accuracy makes it well-suited for locating such structures consistently across diverse patient anatomies and imaging conditions. YOLO models are also lightweight compared to some region-based detectors (e.g., R-CNNs), making them more suitable for real-time clinical deployment, especially in resource-constrained settings. This also facilitates integration into automated diagnostic pipelines. The YOLOv11 pre-trained model was utilized for ROI detection and IVD₁₋₅ identification from axial and sagittal MR images. The process begins with fine-tuning the pre-trained model on a private MRI dataset. The model was then trained separately on sagittal lumbar regions, IVD₁₋₅, and axial IVD₁₋₅ images using a transfer learning approach, where the pre-trained weights served as an initial starting point. This significantly reduced training time and improved performance, particularly when working with smaller datasets. After training, the model was tested on new MRI scans. The identified ROIs were subsequently cropped and analyzed further for disc herniation classification. This approach leverages YOLOv11's high accuracy and scalability to streamline the detection pipeline for clinical applications. The model was optimized using the following hyperparameters: 50 epochs, a batch size of 8, a learning rate (LR) of 0.02 for optimal performance, and an image size of 400×400 .

The lumbar region cropping technique was employed to eliminate extraneous data from the image, focusing solely on the lumbar areas. Otherwise, the model may incorrectly detect IVDs from cervical or other regions, which can degrade the accuracy of lumbar IVD₁₋₅ detection. By removing irrelevant portions of the image, ROI cropping significantly reduces the computational resources required for processing, thereby enhancing the speed and efficiency of the machine learning (ML) models [22]. The YOLOv11 model detects the IVD₁₋₅ with a rectangular bounding box from axial and sagittal images. The area within each bounding box is subsequently cropped. The bounding box dimensions were increased by 10 % to ensure the required IVD information was included. By detecting individual IVD₁₋₅, it is possible to determine whether each lumbar IVD is herniated.

2.4. Classification architectures

The models implemented in this study incorporated a variety of techniques, with a strong emphasis on transfer learning (TL). TL utilizes pre-trained models trained on large-scale datasets to develop robust and generalized neural networks, making it particularly effective for image classification tasks with limited data availability. In this study, attention-based training approaches were applied to fine-tuned CNN models. The architecture of VGG19 is shown in Fig. 3 to illustrate the feature extraction and fine-tuning approaches, integrated with an

attention module [23]. The original top portion of the VGG19 model was replaced with customized layers tailored to the specific classification task. The convolutional and pooling layers of the pre-trained model were imported and frozen, preserving the generalized features learned during pre-training. This method is known as the feature extraction approach. In the fine-tuning approach, select convolutional layers of the pre-trained model were unfrozen to allow retraining. The MRI dataset was passed through the convolutional layers to extract visual features, and only the top few convolutional layers of the pre-trained model were retrained. This selective retraining allowed the model to update the weights of the non-frozen layers based on the axial MRI spine dataset while retaining the weight of the frozen layers. This approach ensured the model adapted effectively to the private MRI dataset while benefiting from the pre-trained network's generalized knowledge. Meanwhile, the newly introduced attention module and fully connected layers were initialized with random weights to adapt effectively to the task-specific dataset. The attention module enabled the model to focus on the most salient features of the input images, thereby significantly enhancing both accuracy and interpretability, particularly for complex image classification tasks. Visualization of attention weights provided insights into the regions of interest in the images, thereby improving the model's transparency. Self-attention mechanisms, such as the SeqSelfAttention layer, were employed to capture long-range dependencies in the input data by attending to all positions within the input sequence [8]. This capability proved beneficial for tasks requiring contextual information from distant regions of the input image. The SeqSelfAttention layer facilitated learning relationships between different regions of the flattened feature map extracted by the VGG19 convolutional base, allowing the model to selectively attend to crucial features for more informed predictions. By focusing on the most relevant features, the attention module improved generalization, robustness, and resistance to overfitting. This ensured the model's adaptability to the private MRI dataset while leveraging the pre-trained network's generalized knowledge. This integration of transfer learning with fine-tuning and attention mechanisms demonstrated a powerful approach to achieving high performance in image classification tasks with limited data.

The primary objective of this stage was to accurately classify axial and sagittal spine MR images into two categories: herniation and no herniation. To achieve this, several Keras CNN architectures were investigated, including,

- Visual Geometry Group Networks (VGG16 and VGG19) [23],
- Residual Networks (ResNet50) [24],
- InceptionV3 [25], and
- DenseNet Connected Convolutional Networks (DenseNet121) [26].

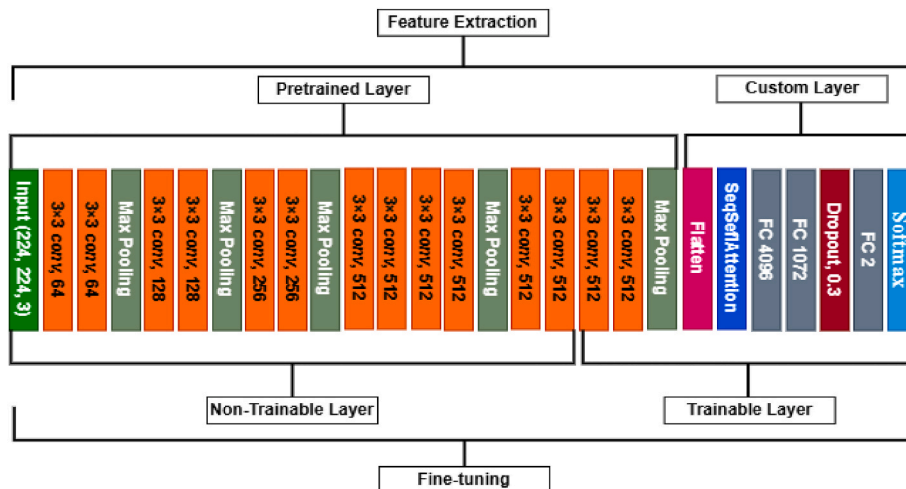


Fig. 3. Feature extraction and fine-tuned VGG19 architecture with attention module.

The VGGNet family, which includes VGG16 and VGG19, is characterized by its structured architecture comprising convolutional, pooling, and fully connected layers. While both variants share similar design principles, a key distinction lies in the number of convolutional layers: VGG19 incorporates fourteen convolutional layers, whereas VGG16 utilizes twelve [23]. This difference in depth reflects a significant variation in the models' capacities to extract features. ResNet, on the other hand, is based on the theory of residual learning, providing a partial solution to the vanishing gradient problem in very deep networks. The architecture includes residual blocks and can reach up to 150 layers in some configurations [24]. The ResNet50 variant was selected for this study due to its superior performance compared to earlier ResNet models. The InceptionV3 model features a more intricate architecture, comprising layers for convolution, pooling, batch normalization, dropout, and fully connected operations [25]. While increasing the depth and complexity of CNN architectures often enhances feature extraction capabilities, it can also lead to diminished information flow from the input to the deeper layers, potentially impacting classification accuracy. DenseNet, on the other hand, addresses this limitation through its innovative design, which incorporates dense blocks. These blocks establish direct connections between all preceding and subsequent layers within each block, effectively improving information flow. The dense blocks in DenseNet are composed of convolutional layers, batch normalization, and ReLU activation functions [26]. This study leveraged these models within a transfer learning framework, using pretrained weights from ImageNet. The transfer learning approach can be particularly beneficial in cases with limited training data [27].

The Keras CNN models were tested in different ways to find the best setup for this classification task. They were evaluated using axial and sagittal ROI images, both with and without fine-tuning, and with and without an attention module. Al-Kubaisi et al. conducted a comparative study of various transfer learning models, with and without axial ROI images and with/without fine-tuning, showing improved accuracy after applying ROI and fine-tuning [15]. Similarly, M.A. Sayed et al. conducted related work on sagittal images using various novel approaches and reached the same conclusion [18]. Therefore, the investigation of full axial/sagittal images (without ROI) and without fine-tuning was excluded, and only fine-tuned approach on ROI images were used in this study. Initially, the individual IVD₁₋₅ discs were detected through the YOLOv11 model, which extracted IVD₁, IVD₂, IVD₃, IVD₄, and IVD₅ regions from axial and sagittal MR images. These axial and sagittal IVD₁₋₅ images were subsequently fed into the classification models for further analysis. Before training, all input images were resized, and the previously described data augmentation techniques were implemented to triple the dataset. The models were trained on both with and without an attention module using the hyperparameter configurations outlined in

Table 1

Summary of hyperparameters used in classification models for axial and sagittal images.

Models	Hyperparameter	Values	Tuned Value
VGG16, Resnet50, InceptionV3, VGG19, and DenseNet121	Number of dropout layers	1, 2	1
	Number of FC layers	1, 2, 3	2
	Dropout Rate	20 %, 30 %, 50 %	50 %
	Epoch	50	Earllystop function
	Batch Size	8, 16, 32, 64	8
	Activation	Softmax, Sigmoid	Softmax
	Optimizer	Adam, SGD	Adam
	Loss Function	Categorical cross-entropy	Categorical cross-entropy
	LR	0.01, 0.001, 0.003	0.001

Table 1, and an early stopping mechanism with a 10-epoch's patience threshold was incorporated into the training process to mitigate overfitting. Model optimization was achieved by systematically adjusting hyperparameters to enhance performance. Additionally, various strategies were used to address class imbalance within the dataset, including data augmentation, algorithmic adjustments, and class weighting. Class weighting is a strategy that assigns a higher weight to the minority class and a lower weight to the majority class. The class weights were determined using sample size weighting, implemented using the *compute_class_weight* function from the *sklearn.utils.class_weight* module. This method calculates weights inversely proportional to class frequencies, using equation (1).

$$Weight_{class} = \frac{\text{total number of images}}{\text{number of classes} \times \text{number of images in that class}} \quad (1)$$

In this study, after augmentation, the number of images in the 'No-PLID' category (majority class) contained 1950 images, while the 'PLID' category (minority class) contained 1344 images. Specific class weights were assigned: 0.85 for the "No-PLID" category and 1.23 for the "PLID" category.

After training, evaluation metrics were computed to assess model performance. By integrating the detection of IVD₁₋₅ using the YOLOv11 model with the classification of the IVDs, it became possible to determine which lumbar IVDs were herniated. The axial and sagittal IVD₁₋₅ classification workflow is shown schematically in Fig. 4.

2.5. Probability calculations

In the process, a deep learning approach based on the attention-based, fine-tuned VGG19 and ResNet50 architectures was employed, generating raw prediction scores for each class. These scores were converted into class probabilities using the softmax activation function, ensuring the predicted values were interpretable as probabilities.

The model processes the input image and generates raw output logits (z_i), where each element z_i represents the model's confidence score for class 'i'. These logits are transformed into class probabilities using the softmax function:

$$P(y_i) = \frac{e^{z_i}}{\sum_{j=1}^N e^{z_j}} \quad (2)$$

where $P(y_i)$ is the probability of class, z_i is the raw score for class, and N is the total number of classes (here, $N = 2$ for PLID and No-PLID).

After applying the softmax function, a probability distribution over the two classes was obtained. The model assigns a percentage confidence to each class, allowing for an intuitive interpretation of the result.

2.5.1. Decision rule for the final classification

Since attention-based, fine-tuned VGG19 and ResNet50 architectures were used for axial and sagittal images, respectively, the final classification decision for a patient's disc herniation or not was based on the following rules:

Case 1. If both VGG19 and ResNet50 predict the same class (either PLID or No-PLID), the final decision was directly taken as PLID or No-PLID. This scenario represents high-confidence predictions where both models independently validate the same outcome.

Case 2. If the models disagree (one predicts PLID, the other predicts No-PLID), the decision was made based on the higher probability score assigned by either model.

If both models (VGG19 and ResNet50) agreed on either PLID or No-PLID, the final classification was based on their consensus. Otherwise, the decision was made by the model that assigns a higher probability to its classification choice. Fig. 5 illustrates the flowchart of combining decisions from the VGG19 and ResNet50 models for disc herniation

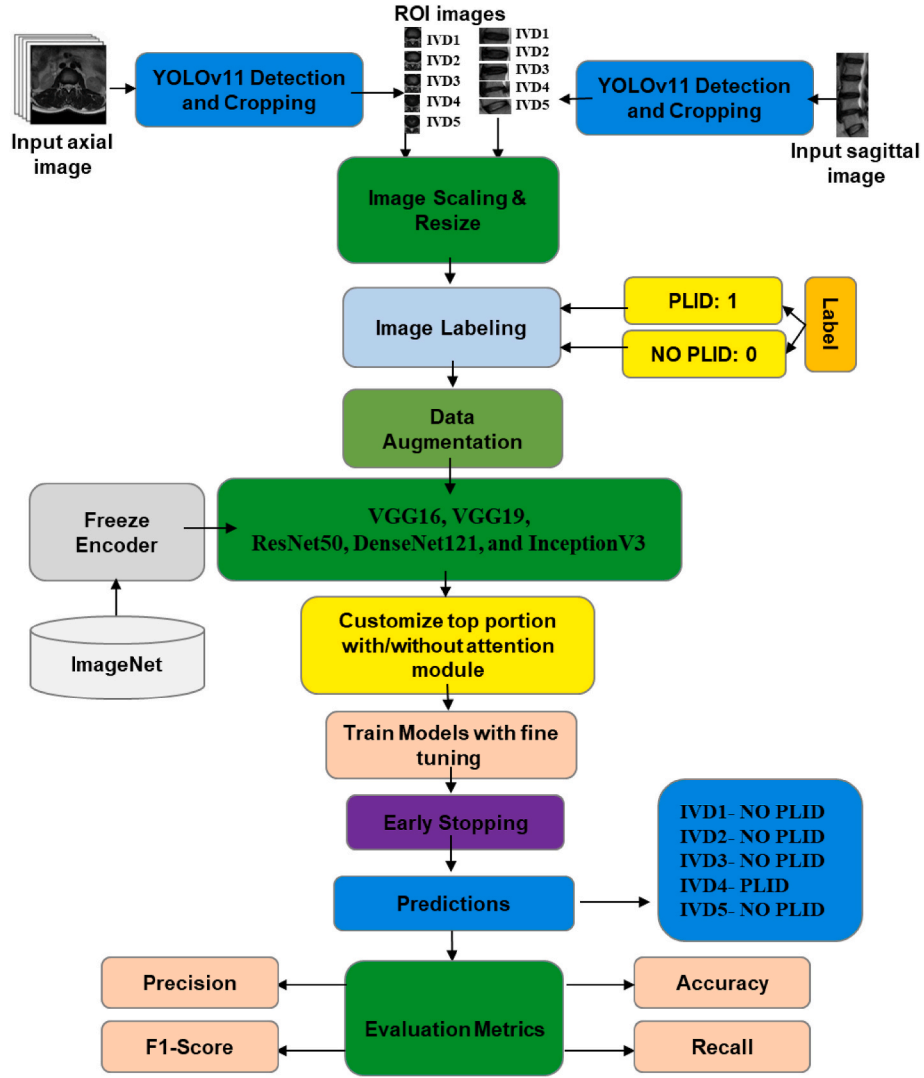


Fig. 4. Workflow for the classification of IVD₁₋₅ in axial and sagittal MR images, investigating Keras models with and without attention module.

classification in a patient.

- PLID_a represents the probability value of PLID predicted by the VGG19 model from axial MR images.
- PLID_s represents the probability value of PLID predicted by the ResNet50 model from sagittal MR images.
- PLID_p is the final classification as PLID for a patient, determined by integrating the decisions from both models using axial and sagittal MR images.

2.6. Evaluation metrics

Evaluation metrics are essential for measuring a model's performance on a dataset. They help measure the effectiveness of the classifier during both the training and testing phases. It is important to use multiple evaluation metrics, as a model's performance can vary depending on the specific criterion. A classifier may perform well on some metrics while underperforming on others. The key evaluation metrics used in this study are described below.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (3)$$

$$Specificity = \frac{TP}{TP + FP} \quad (4)$$

$$Recall = \frac{TP}{TP + FN} \quad (5)$$

$$Precision = \frac{TP}{TP + FP} \quad (6)$$

$$F1 - Score = \frac{2(Precision * Recall)}{(Precision + Recall)} \quad (7)$$

$$Average\ precision\ (AP) = \sum_{i=1}^{N-1} (Recall_{i+1} - Recall_i) * Precision_{i+1} \quad (8)$$

$$Mean\ average\ precision\ (mAP) = \frac{1}{M} \sum_{i=1}^n AP_i \quad (9)$$

where TP = true positives, TN = true negatives, FP = false positives, and FN = false negatives. N represents the total number of recall points, and M represents the total number of classes.

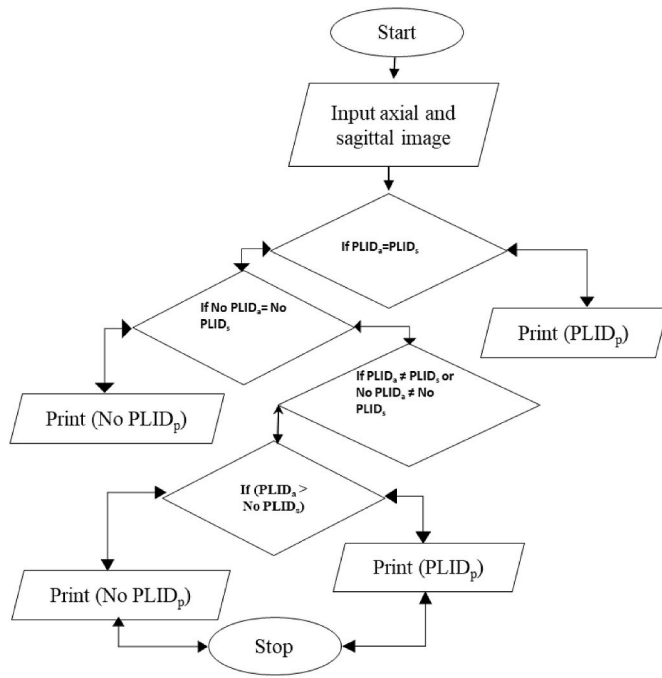


Fig. 5. Flowchart illustrating the decision fusion process between the VGG19 and ResNet50 models.

3. Results and discussion

3.1. Detection of ROI

The model was trained and validated using axial and sagittal MR images from the dataset. As the metrics shown in Fig. 6 indicate, the mean average precision at IoU 0.50 (mAP50) increased steadily in epoch increments, indicating that the training process was working as intended. The training mAP50 curves reached 99.50 %, while mAP50-95 curves reached up to 92 %, which shows that the model can detect the ROI well. It was calculated to measure the model's performance, showing a sharp increase in accuracy, confirming successful training progression.

The parameters used for model performance evaluation include mAP50 and mAP50-95. Among these, mAP50 represents the mAP calculated at a fixed IoU threshold of 0.5. In contrast, mAP50-95 provides a more comprehensive assessment by averaging the AP values computed across multiple IoU thresholds, ranging from 0.5 to 0.95. These metrics offer a detailed understanding of the model's detection and localization capabilities. The specific results obtained from the ROI detection are presented in Table 2. The YOLOv11 model achieved

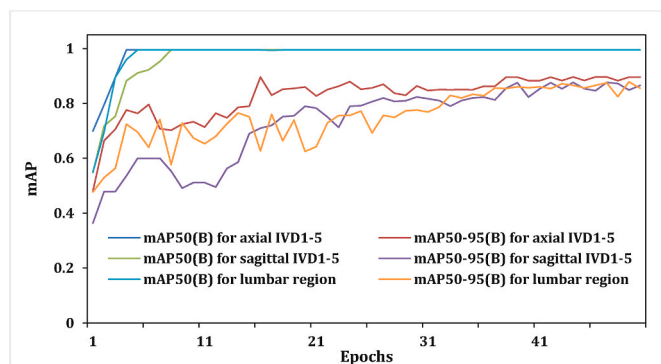


Fig. 6. Training mAP50 and mAP50-95 curves of the YOLOv11 model for detecting lumbar regions, axial and sagittal IVD₁₋₅.

Table 2

Performance of the YOLOv11 model for detecting lumbar regions, axial and sagittal IVDs model from MR images.

Image view	mAP50 for IVD ₁₋₅ (%)	Average mAP50 (%)	mAP50-95 for IVD ₁₋₅ (%)	Average mAP50-95 (%)
Axial IVD	92.33 96.75 91.50 99.50 95.80	95.18	87.62 89.50 86.85 91.25 85.78	88.20
Sagittal lumbar	–	99.50	–	89.50
Sagittal IVD	95.50 83.33 97.70 99.50 98.33	94.87	77.50 85.58 79.60 82.36 75.36	80.08

mAP50 scores of 95.18 %, 99.50 %, and 94.87 %, and mAP50-95 scores of 88.20 %, 89.50 %, and 80.08 % for axial IVD₁₋₅, lumbar regions, and sagittal IVD₁₋₅, respectively. This indicates that the mAP50 scores were higher than the mAP50-95 scores because mAP50 is calculated at a single IoU threshold, whereas mAP50-95 accounts for a range of thresholds. For ROI cropping, an IoU score of 0.25 was used to ensure 100 % detection accuracy. YOLO's ability to detect multiple ROIs with high spatial accuracy likely contributed to its superior accuracy and speed compared to models like Faster R-CNN, SSD, DETR, and EfficientDet [28–30].

The detection and cropping outcomes for each axial and sagittal IVD₁₋₅ are illustrated in Fig. 7. In Fig. 7a, the YOLOv11 model performed axial IVD₁₋₅ detection, identifying the individual IVDs. The detected IVDs were then cropped to create the ROI images, which were subsequently fed into the next classification model. In Fig. 7b, the model detected and cropped the lumbar region to ensure that only the lumbar area was considered. This prevented the model from detecting errors in IVDs from the cervical or other regions, which could negatively impact lumbar IVD₁₋₅ detection results. In Fig. 7c, the model detected and cropped sagittal IVD₁₋₅, creating ROI images that were then fed into the classification model.

3.2. Classification of IVDs

In the study, the comparison between different models with an attention module on axial and sagittal images was employed. Various training procedures were employed with deep learning CNN models (VGG16, VGG19, DenseNet121, ResNet50, and InceptionV3) to classify the disc state (No-PLID and PLID). Finally, an attention-based fine-tuned VGG19 and ResNet50 model for axial and sagittal MR images were introduced, respectively, to show how the new method performs better than each of the other models and procedures. From the accuracy vs epoch curve for axial image in Fig. 8a, the key observation was that the training and validation accuracy increased with the increment of epochs and fluctuated in the later epochs. This fluctuation was due to variations in the training data. The VGG19 model achieved the highest training and validation accuracy. Therefore, it can be concluded that the proposed model has performed well during the training and validation phases. From the accuracy vs epoch curve for sagittal image in Fig. 8b, it was observed that the percentage of training and validation accuracy increased sharply in the initial epochs. The sharp increase observed during the initial epochs indicates that the model rapidly improved in performance during the early stages of training (i.e., the first few passes through the dataset). This can be attributed to the use of pretrained models, which had already learned general features such as edges, textures, and shapes from large-scale datasets like ImageNet. When fine-tuned on the MRI dataset, these models started from pretrained weights and quickly adapted to the new domain by adjusting the higher-level layers. As a result, both training and validation accuracy improved

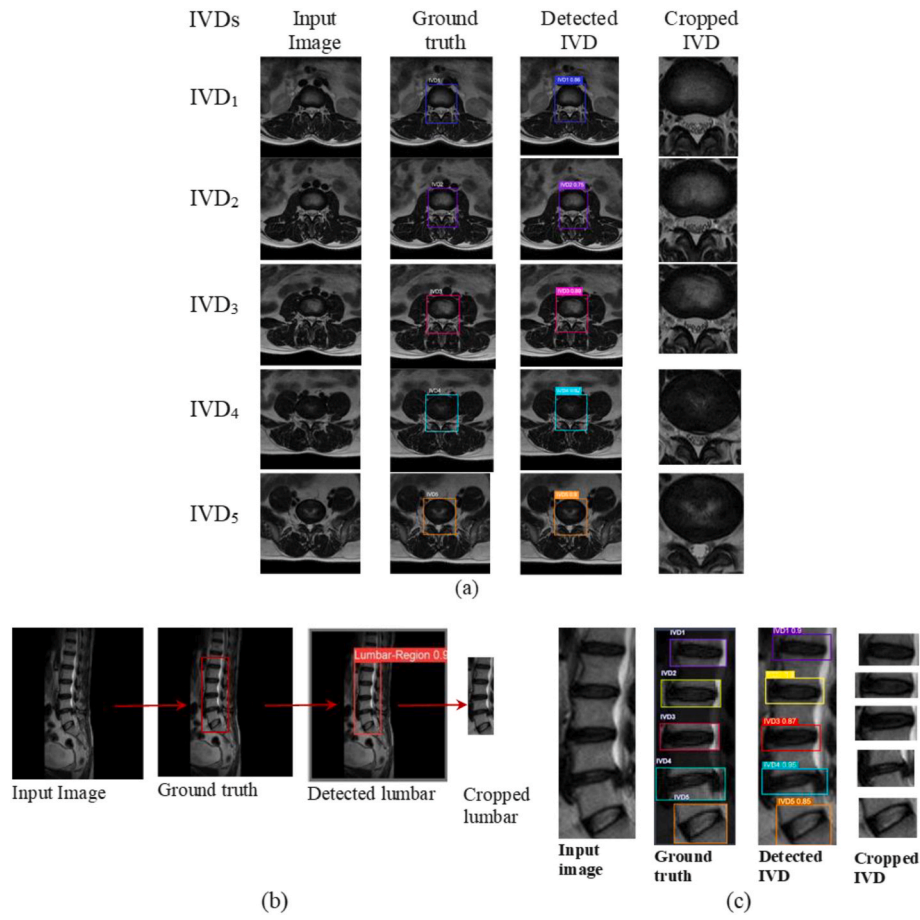


Fig. 7. Detection and cropping outcomes for (a) axial IVD₁₋₅ (b) the sagittal lumbar region and (c) the sagittal IVD₁₋₅.

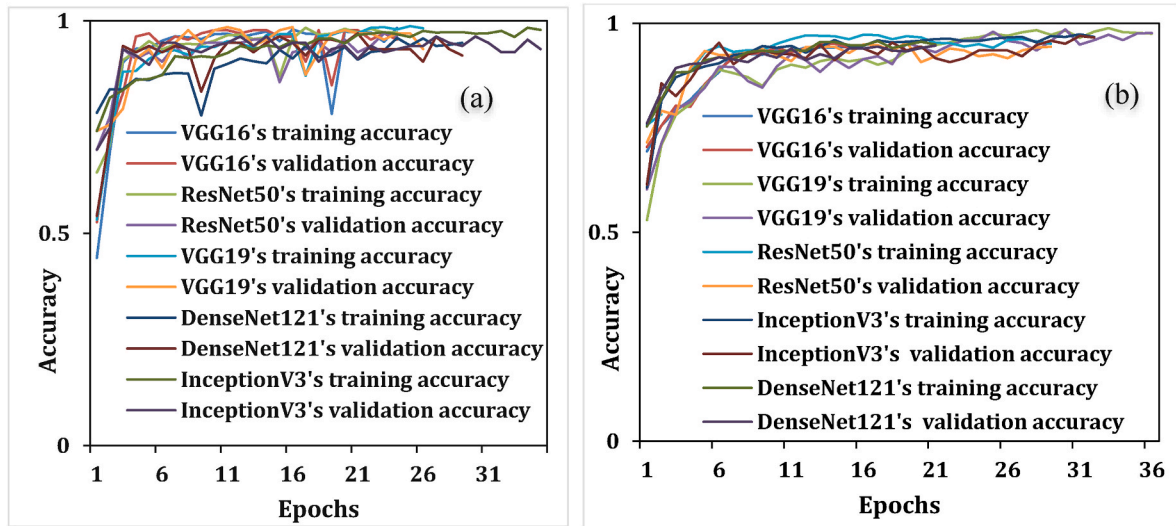


Fig. 8. Training and validation accuracy curves for fine-tuned CNN models with attention module on (a) axial and (b) sagittal images.

significantly during the initial training phase. The ResNet50 model achieved the highest training and validation accuracy among other models, making it the recommended model for sagittal IVD classifications.

The confusion matrix and receiver operating characteristic (ROC) curve of the attention-based fine-tuned VGG19 and ResNet50 models for classifying disc herniation from axial and sagittal MR images are

visualized in Fig. 9. The VGG19 demonstrates excellent performance for axial images, accurately classifying 97 % of No-PLID images and 97.14 % of PLID images (shown in Fig. 9a). The ResNet50 model accurately classified 98.20 % of No-PLID images and 95.65 % of PLID images (shown in Fig. 9b).

This high level of accuracy indicates the model's robustness in distinguishing between the two classes, effectively capturing the subtle

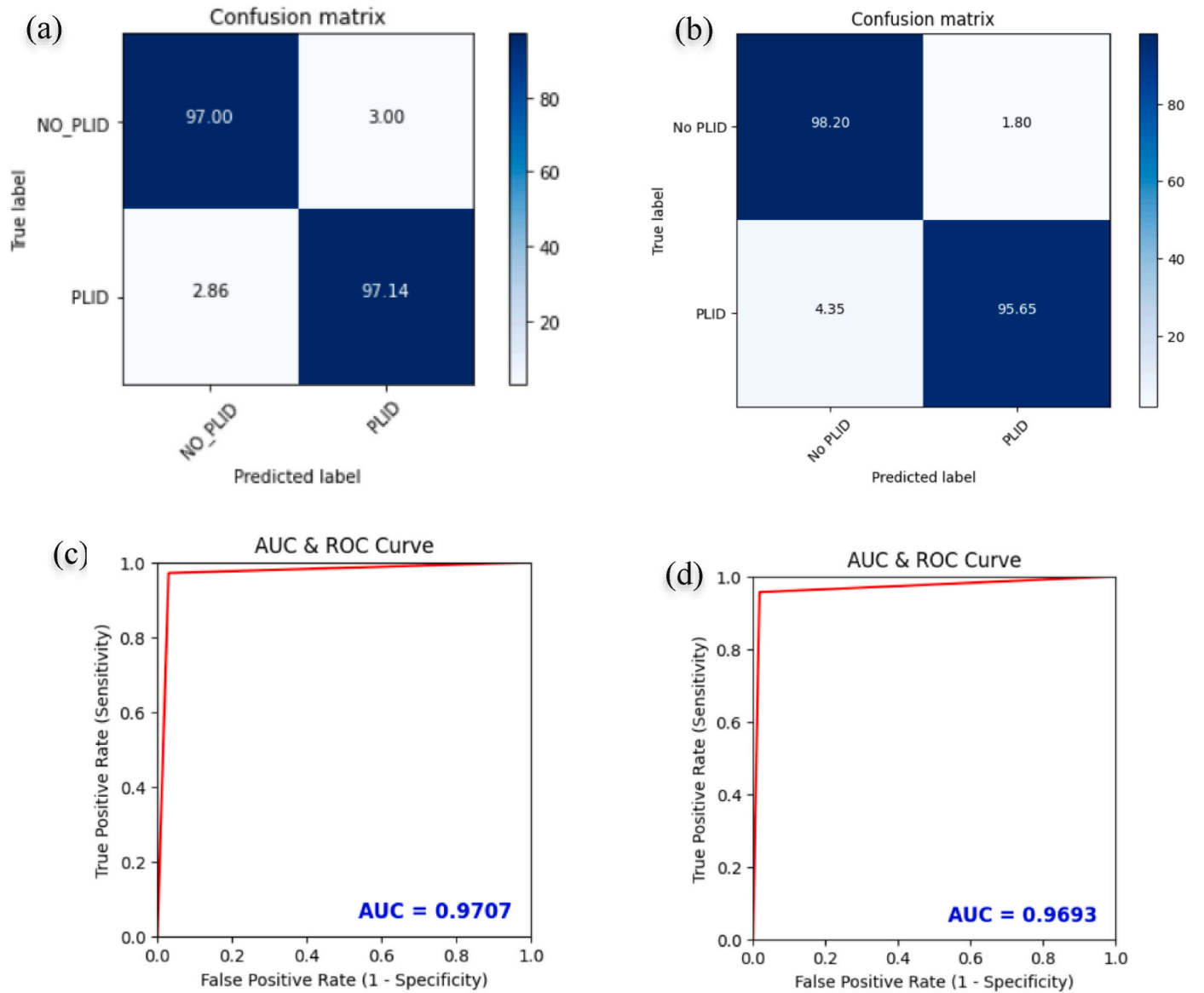


Fig. 9. Confusion matrices for the attention-based fine-tuned models: (a) VGG19 on axial MR images and (b) ResNet50 on sagittal MR images. ROC curves for (c) VGG19 on axial MR images and (d) ResNet50 on sagittal MR images.

differences between No-PLID and PLID cases. Despite minor classification errors, the models maintain strong diagnostic performance, as evidenced by the AUC of 97.07 % for VGG19 and 96.93 % for ResNet50, depicted in Fig. 9c–d. This high AUC score reflects the model's ability to discriminate between No-PLID and PLID cases with exceptional precision and reliability.

In Grad-CAM, heat maps were generated from the final convolutional layer of the model. These heat maps were designed to highlight the regions that provide the most accurate visual representation of the object based on the model's interpretation. As illustrated in Fig. 10, Grad-CAM visualizations of features using the attention-based fine-tuned VGG19 and ResNet50 models were employed for disc state classification. After implementing the ROI technique and attention module, the models demonstrated a clear focus on the herniated regions, along with the extraction of features from the surrounding abnormal tissue. The reddish areas in the heat maps indicate regions where the model's attention was most concentrated.

Table 3 highlights that integrating an attention module significantly enhances classification performance in deep learning models for axial spine ROI MR images. Among the models, the fine-tuned VGG19 with an

attention module achieved the highest accuracy (97.05 %), making it the recommended model. ResNet50 and InceptionV3 also showed notable improvements of 2.50 % and 2.32 %, respectively, while DenseNet121 showed only minimal benefit (an improvement of 0.11 %). The attention module helps models focus on relevant features, increasing accuracy, AUC, recall, precision, and F1-score. Overall, attention-enhanced models, particularly VGG19, demonstrate superior classification performance, reinforcing the importance of attention mechanisms in disc herniation analysis.

Table 4 shows the performance of the models on sagittal spine ROI MR Images after applying fine-tuning and the attention module. ResNet50 achieved the highest accuracy (97.45 %), showing an increase compared to its performance on axial MR images. VGG19 maintained strong performance with an accuracy of 95.14 %, while InceptionV3 (94.86 %) and DenseNet121 (93.78 %) also demonstrated considerable improvements. The results suggest that sagittal MR images benefit significantly from fine-tuning and attention mechanisms, particularly with ResNet50, which maintains a balance between recall (95.65 %) and precision (95.65 %). These results underline the effectiveness of ResNet50 in handling sagittal images.

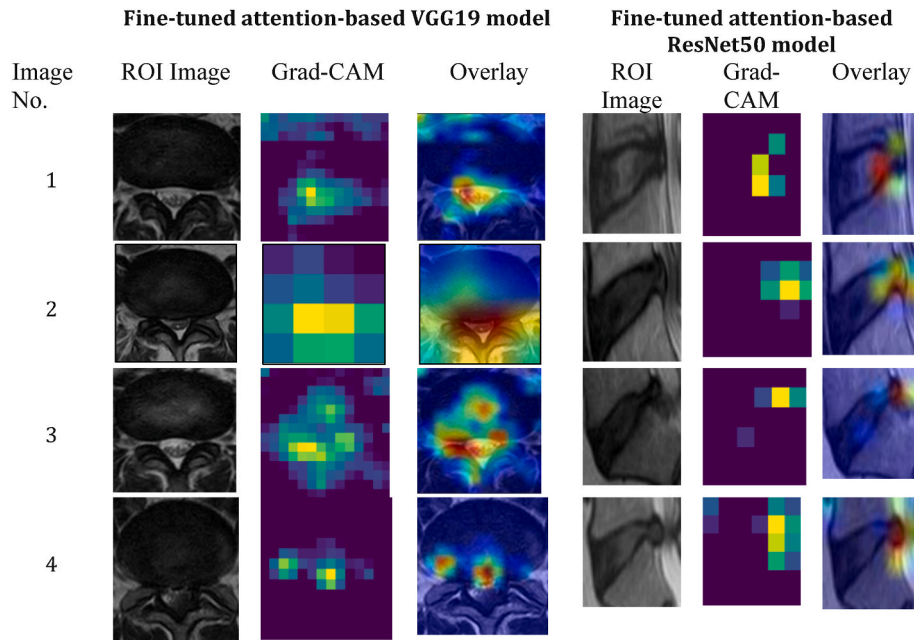


Fig. 10. Grad-CAM visualizations of features extracted by the fine-tuned attention-based VGG19 and ResNet50 models after applying the ROI technique.

Table 3

Performance of the models with fine-tuning, with and without attention module on axial spine ROI MR image.

Models	Accuracy without attention module (%)	Accuracy with attention module (%)	AUC (%)	Recall (%)	Precision (%)	F1-Score (%)
VGG16	93.53	95.47	94.57	93.14	92.44	92.79
VGG19	96.12	97.05	97.07	97.14	95.77	96.45
ResNet50	93.99	96.49	96.14	94.29	97.06	95.65
DenseNet121	94.01	94.12	93.93	92.86	92.86	92.86
InceptionV3	92.19	94.51	94.64	94.28	92.96	93.62

Table 4

Performance of the models with fine-tuning and attention module on sagittal spine ROI MR image.

Models	Accuracy (%)	AUC (%)	Recall (%)	Precision (%)	F1-Score (%)
VGG16	93.59	92.15	95.31	89.39	92.27
VGG19	95.14	94.59	91.36	96.78	93.94
ResNet50	97.45	96.93	95.65	95.65	95.65
DenseNet121	93.78	92.95	91.66	90.75	91.20
InceptionV3	94.86	92.82	93.95	91.14	92.58

Table 5 presents the performance of models tested on the test dataset, including axial and sagittal MR ROI images. VGG19 and ResNet50 maintained competitive accuracy levels (97.05 % and 97.45 %, respectively). However, the most significant improvement was observed with the combined decision from the models, which achieved an outstanding accuracy of 98.09 %. The recall (97.83 %), precision (95.75 %), and F1-score (96.78 %) further support the efficacy of the combined decision-making. This result highlights that combining the VGG19 and ResNet50 models' decisions enhances classification performance, outperforming individual models.

Table 5

Performance of the models with fine-tuning and attention module on axial and sagittal spine ROI MR image.

Models	Accuracy	Recall	Precision	F1-Score
VGG19 (for axial image)	97.05	97.14	95.77	96.45
ResNet50 (for sagittal image)	97.45	95.65	95.65	95.65
Combining decisions from both classification models (proposed method)	98.09	97.83	95.75	96.78

%, and F1-score (96.78 %) further support the efficacy of the combined decision-making. This result highlights that combining the VGG19 and ResNet50 models' decisions enhances classification performance, outperforming individual models.

The results across the tables indicate that fine-tuning and attention modules significantly improve classification performance on spine ROI MR images. VGG19 and ResNet50 consistently perform well across different datasets, with VGG19 excelling in axial images and ResNet50 in sagittal images. The findings suggest that integrating attention mechanisms and fine-tuning strategies are crucial for achieving superior diagnostic accuracy in MR image classification. Both VGG19 and ResNet50 were selected for MR image analysis due to their advanced architectures and transfer learning potential, which outperform basic CNNs (such as AlexNet, MobileNet, SqueezeNet, EfficientNet, NASNet, etc.) [31,32]. Moreover, they are more appropriate than RNNs (LSTM, GRU, etc.) for processing spatial data. Unlike RNNs, which are designed for sequential inputs, VGG19 and ResNet50 are optimized for image-based tasks [33]. Their selection was supported by superior performance across multiple evaluation metrics, confirming their effectiveness for intervertebral disc herniation classification. VGG19 performed best for axial MR images, possibly due to its balanced architecture with 14 convolutional layers, meaning it is neither too shallow nor too deep compared to other models. This balance enables it to effectively capture fine-grained, structured features typical in axial slices. In contrast, ResNet50 excelled with sagittal MR images due to its residual learning architecture, which mitigates the vanishing gradient problem and ensures smooth information flow across deeper layers [34]. This capability made ResNet50 ideal for handling the complex, less structured nature of sagittal images, leading to superior performance in

feature extraction and generalization. Fig. 11 presents a graphical representation of various strategies, comparing the accuracy of fine-tuning approaches with and without the attention module. The results demonstrate increased accuracy when the attention module is applied to ROI images.

The combination of both models further enhances accuracy, demonstrating the potential of combining decisions from both classification models in PLID analysis from axial and sagittal MR images (shown in Fig. 12). If both models (VGG19 and ResNet50) agree on PLID or No-PLID, the final decision for the patient is taken PLID or No-PLID. Otherwise, the decision is based on the probability assigned by the model that gives a higher value to its choice. For IVD₁ of patient 1, the VGG19 model predicted a PLID_a image as No-PLID_a with a 65.35 % probability and the ResNet50 model predicted a PLID_s image as PLID_s with a 75.36 % probability. As the condition $PLID_s > No-PLID_a$ holds, the final decision is PLID_p. For IVD₅ of patient 2, the VGG19 model predicted a PLID_a image as No-PLID_a with a 79.25 % probability and the ResNet50 model predicted a PLID_s image as No-PLID_s with a 68.25 % probability. As condition $PLID_a > No-PLID_s$ holds, the final decision is PLID_p.

Axial and sagittal images capture different features relevant to disc herniation detection. In some cases, disc herniation is more prominently observed in the axial view than in the sagittal view, and vice versa. If a herniation was present but misclassified as non-herniated, the classification probability was often lower than the actual probability, eventually leading the model to classify it as herniated. This occurs because the features predicting herniation do not fully resemble those of a non-herniated disc. Both models have distinct architectures and feature extraction capabilities. VGG19 captures finer texture details, while ResNet50 excels in deeper hierarchical feature extraction. By combining their outputs, a more robust and accurate decision-making process was achieved. Instead of relying on a single model, the classification decision in cases of disagreement was made by the model with the higher confidence score, leading to improved reliability in final classification. This fusion mechanism helps mitigate biases from individual models and MRI scan views (axial or sagittal), ensuring better adaptability across diverse datasets. Overall, combining decisions from both classification models significantly enhances the accuracy of disc herniation detection for a patient.

Finally, the attention-based fine-tuned VGG19 and ResNet50 models achieved better results for disc state classification with the ROI approach. After combining decisions from both models, the classification accuracy further increased. This can be seen in Table 5, where the model accuracy and other metrics achieved by the proposed method represent the best results. The results are compared with previous studies conducted by different researchers in Table 6. The proposed method outperforms existing models in accuracy, 98.09 %, and recall, 97.83 %, with a precision of 95.75 % and F1-score of 96.78 %. In contrast, VGG16 used by A. Kubaisi et al. achieved an 89.01 % accuracy, and ResNet101 used by Q. Pan et al. reached 88.76 %, both significantly

lower. Although M. Ebrahim et al. reported a high accuracy of 98 % using SVM with dual feature extractors (VGG19 and ResNet101), their dataset was notably smaller (164 images), which limits the generalizability of their findings. Other approaches, such as DiscNet, AlexNet, and CMMF-Net, demonstrated relatively lower performance, particularly in precision and F1-score. The attention module in the proposed model improved its ability to focus on relevant features in MR images. Additionally, the combined decision-making from axial and sagittal views using VGG19 and ResNet50, respectively, leveraged the strengths of both perspectives for disc herniation identification, ensuring a more comprehensive and accurate analysis.

4. Conclusions

This study presents a robust deep-learning-based approach for the detection and classification of lumbar disc herniation, integrating advanced techniques such as fine-tuning, attention modules, and automated ROI extraction. By leveraging YOLO for precise lumbar IVD₁₋₅ detection and combining decisions from attention-based fine-tuned VGG19 and ResNet50 models based on their probability scores for classification of disc herniation in a patient's IVDs, the proposed method significantly enhances diagnostic accuracy. The method demonstrated superior performance on the test dataset for disc herniation classifications in a patient, achieving an accuracy of 98.09 %, a recall of 97.83 %, a precision of 95.75 %, and an F1-score of 96.78 %, outperforming existing approaches. Additionally, Grad-CAM visualizations validated the model's interpretability by highlighting critical regions contributing to classification, making it more reliable for clinical applications. While dataset limitations posed a challenge, implementing data augmentation, dropout layers, early stopping, and transfer learning effectively mitigated overfitting and improved generalizability.

This framework offers significant potential in clinical settings by automating lumbar disc herniation detection and classification, reducing radiologists' workload while ensuring high diagnostic accuracy and consistency. The proposed deep learning-based intervertebral disc (IVD) classification system can be deployed in real-time through an integrated hardware-software solution suitable for clinical environments. Trained models like VGG19 (for axial) and ResNet50 (for sagittal images) can be embedded into compact devices such as the NVIDIA Jetson or a Raspberry Pi with TPU. Alternatively, the system can be implemented as a web application (using Django or Flask framework) or a mobile application, which can be used on a mobile device or PCs [16]. These applications would allow radiologists or physicians to upload MR images directly into the system, automatically preprocess the images, perform classification using trained deep learning models, and present the diagnostic output within seconds. To enable real-time performance, the trained models can be deployed using lightweight frameworks such as TensorFlow Lite or TensorFlow.js. These frameworks are optimized for mobile and web platforms, enabling fast inference with minimal computational resources. The system can be hosted on a secure cloud

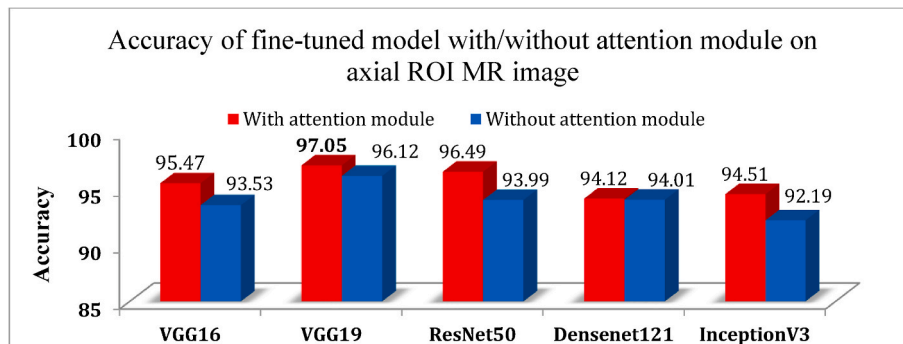


Fig. 11. Comparison of the classification accuracy of the fine-tuning approach with and without attention module for axial MR images.

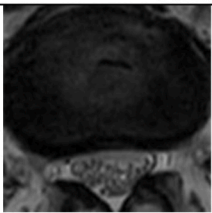
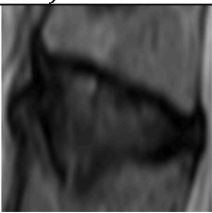
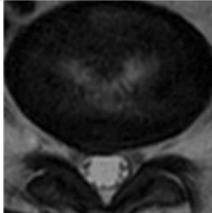
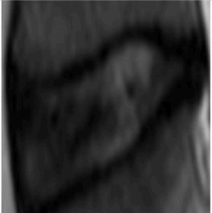
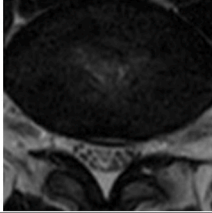
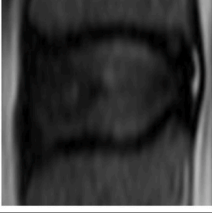
Patient	IVD	VGG19	ResNet50	Probability comparison (Pseudo code)	Decision
		PLID(actual)			
1	IVD ₁			If PLID _a = PLID _s Print (PLID _p) Elseif No-PLID _a = No-PLID _s Print (No-PLID _p) Elseif PLID _a ≠ PLID _s or No-PLID _a ≠ No-PLID _s If PLID _a > No-PLID _s or PLID _s > No-PLID _a Print(PLID _p) Else Print(No-PLID _p)	PLID _p
		No-PLID _a - 65.35% (predicted)	PLID _s - 75.36% (predicted)		
2	IVD ₅			If PLID _a = PLID _s Print (PLID _p) Elseif No-PLID _a = No-PLID _s Print (No-PLID _p) Elseif PLID _a ≠ PLID _s or No-PLID _a ≠ No-PLID _s If PLID _a > No-PLID _s or PLID _s > No-PLID _a Print(PLID _p) Else Print(No-PLID _p)	PLID _p
		PLID _a - 79.25% (predicted)	No-PLID _s - 68.2% (predicted)		
3	IVD ₄			If PLID _a = PLID _s Print (PLID _p) Elseif No-PLID _a = No-PLID _s Print (No-PLID _p) Elseif PLID _a ≠ PLID _s or No-PLID _a ≠ No-PLID _s If PLID _a > No-PLID _s or PLID _s > No-PLID _a Print(PLID _p) Else Print(No-PLID _p)	PLID _p
		PLID _a - 85.15% (predicted)	PLID _s - 80.17% (predicted)	Print(PLID _p) Else Print(No-PLID _p)	

Fig. 12. Illustration of the decision making process in disc herniation classification.

Table 6

Comparison of the proposed method with previous works.

Author	Model	Image plane	Image number	Accuracy	Recall	Precision	F1-Score
Our proposed method	Attention –based fine-tuned VGG19 and ResNet50 (combine decision)	axial & sagittal	1098 (axial) 1098 (sagittal)	98.09	97.83	95.75	96.78
A. kubaisi et al. [15]	VGG16	axial	905	89.01	90.91	90.91	90.91
Q. Pan et al. [16]	ResNet101	axial	3555	88.76	–	–	–
M. Alsmirat et al. [17]	AlexNet	axial	164	91.38	–	–	–
T. Šušteršič et al. [35]	DiscNet	axial	557	90.00	86.25	86.25	–
		sagittal	1169	91.00	91.00	91.67	–
E. Salehi et al. [36]	AlexNet	axial	2329	87.75	–	–	–
S. Lu et al. [37]	CMMF-Net	axial	1569	91.32	83.96	–	–
M. Ebrahim et al. [38]	VGG19 & ResNet101 with SVM	axial	164	98.00	–	–	–
D.S. Tan et al. [39]	NLID	sagittal	98	96.88	–	–	96.46

platform (e.g., AWS, Google Cloud, or Microsoft Azure) for web-based applications. Users would access the service through a browser, and the backend server would handle image classification and display the result with diagnostic recommendations. For mobile applications, the model can be embedded within the app or connected to a cloud API for inference. Mobile apps offer portability, enabling clinicians to conduct assessments at the bedside, in outpatient settings, or even in resource-constrained environments.

Future research can focus on expanding the dataset by incorporating images from diverse sources and different MRI devices to enhance model robustness further. Moreover, integrating self-supervised learning techniques and exploring 3D imaging could optimize the detection and

classification of complex spinal pathologies, paving the way for more advanced AI-driven diagnostic systems in medical imaging.

CRediT authorship contribution statement

Md Abu Sayed: Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Ashiqur Rahman:** Writing – review & editing, Formal analysis, Data curation. **Sadman Mohammad Nasif:** Writing – review & editing, Formal analysis. **Sudipto Halder:** Writing – review & editing, Formal analysis. **Akram Hossain:** Writing – review & editing, Formal analysis, Data curation. **Hasan Ahmed:** Writing – review & editing, Formal analysis. **Muhammad Abdul Kadir:** Writing –

review & editing, Visualization, Supervision.

Data availability

The dataset was obtained upon request from the corresponding author of the paper [18]. Additional data is available upon request from the corresponding author.

Ethics statement

The MR image dataset was obtained upon request from the corresponding author of the paper [18], which is fully anonymized and complies with all relevant ethical standards for the collection and distribution of human subject data.

Funding

The author MAK acknowledges the International Science Programme (ISP) of Uppsala University in Sweden for financial support as a research grant.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

The authors gratefully thank Md. Sherajul Islam of Khulna University of Engineering and Technology, Bangladesh, for providing the dataset.

References

- [1] Ferreira ML, de Luca K, Haile LM, Steinmetz JD, Culbreth GT, Cross M, et al. Global, regional, and national burden of low back pain, 1990–2020, its attributable risk factors, and projections to 2050: a systematic analysis of the Global Burden of Disease Study 2021. *Lancet Rheumatol* 2023;5(6):e316–29. [https://doi.org/10.1016/s2665-9913\(23\)00098-x](https://doi.org/10.1016/s2665-9913(23)00098-x).
- [2] Anselmo A, Pagano M, Cappadona I, Cardile D, Russo F, Laudisio A, et al. A systematic review on the neuropsychological assessment of patients with LBP: the impact of chronic pain on quality of life. *J Clin Med* 2024;13(20):6149. <https://doi.org/10.3390/jcm13206149>.
- [3] Rahman H, Haque A, Emran M, Zaman A. Conservative treatment alone and with diclofenac sodium phonophoresis on the patients with low back pain due to prolapsed lumbar intervertebral disc. *KYAMC J* 2022;13:158–64. <https://doi.org/10.3329/kyamcj.v13i3.63204>.
- [4] Zhang AS, Xu A, Ansari K, Hardacker K, Anderson GM, Alsoof D, et al. Lumbar disc herniation: diagnosis and management. *Am J Med* 2023;136(7):645–51. <https://doi.org/10.1016/j.amjmed.2023.03.024>.
- [5] Alanazi T, Al-Anzi A, Alsafari A, Enezi A, Alanazi F, Alanazi M, et al. Updates in the role of MRI in diagnosis of lumbar disc herniation: a systematic review article. *J Adv Med Med Res* 2022;34:22–30. <https://doi.org/10.9734/jammr/2022/v34i244900>.
- [6] Alexander K. Reducing error in radiographic interpretation. *Can Vet J* 2010;51(5): 533–6. <https://pubmed.ncbi.nlm.nih.gov/20676300/>.
- [7] Aboulmira A, Raouhi EM, Hrimech H, Lachgar M. Ensemble learning methods for deep learning: application to skin lesions classification. 11th Intern Sympos Signal, Image, Video and Commun 2022:1–6. <https://doi.org/10.1109/ISIVC54825.2022.9800732>.
- [8] Vaswani A, Shazeer NM, Parmar N, Uszkoreit J, Jones L, Gomez AN, et al. Attention is all you need. In: Proceedings of the 31st international conference on neural information processing systems (NIPS'17). NY: Curran Associates Inc; 2017. p. 6000–10. <https://doi.org/10.48550/arXiv.1706.03762>.
- [9] Apostolopoulos ID, Papathanasiou ND, Papandrianos N, Papageorgiou E, Apostolopoulos DJ. Innovative attention-based explainable feature-fusion VGG19 network for characterising myocardial perfusion imaging SPECT polar maps in patients with suspected coronary artery disease. *Appl Sci* 2023;13:8839. <https://doi.org/10.3390/app13158839>.
- [10] Ebrahimzadeh E, Fayaz F, Ahmadi F, Nikravan M. A machine learning-based method in order to diagnose lumbar disc herniation disease by MR image processing. *MedLife Open Access* 2018;1(1):1–10. Available: <https://www.medtextpublications.com/medlife/articles/MOA-0002.pdf>. [Accessed 31 March 2025].
- [11] Unal Y, Polat K, Kocer HE, Hariharan M. Detection of abnormalities in lumbar discs from clinical lumbar MRI with hybrid models. *Appl Soft Comput* 2015;33:65–76. <https://doi.org/10.1016/j.asoc.2015.04.031>.
- [12] Hou C, Li X, Wang H, Zhang W, Liu F, Liu D, et al. An MRI image automatic diagnosis model for lumbar disc herniation using semi-supervised learning. *Complex Intelligent Syste* 2023;9(5):5567–84. <https://doi.org/10.1007/s40747-023-00981-0>.
- [13] Ghosh S, Raja SA, Chaudhary V, Dhillon G. Computer-aided diagnosis for lumbar mri using heterogeneous classifiers. *IEEE international symposium on biomedical imaging: from nano to macro*. 2011. p. 1179–82. <https://doi.org/10.1109/ISBI.2011.5872612>.
- [14] Mbarki W, Bouchouicha M, Frizzi S, Tshibasu F, Farhat LB, Sayadi M. Lumbar spine discs classification based on deep convolutional neural networks using axial view MRI. *Interdiscip Neurosurg* 2020;22:100837. <https://doi.org/10.1016/j.inat.2020.100837>.
- [15] Al-kubaisi A, Khamiss NN. A transfer learning approach for lumbar spine disc state classification. *Electronics* 2021;11(1):85. <https://doi.org/10.3390/electronics11010085>.
- [16] Pan Q, Zhang K, He L, Dong Z, Zhang L, Wu X, et al. Automatically diagnosing disk bulge and disk herniation with lumbar magnetic resonance images by using deep convolutional neural networks: method development study. *JMIR Med Inform* 2021;9(5):e14755. <https://doi.org/10.2196/14755>.
- [17] Alsmirat M, Al-Mnayyis N, Al-Ayyoub M, Al-Mnayyis A. Deep learning-based disk herniation computer aided diagnosis system from MRI axial scans. *IEEE Access* 2022;10:32315–23. <https://doi.org/10.1109/ACCESS.2022.3158682>.
- [18] Sayed MA, Rahman GMM, Islam MS, Islam MA, Park J, Ahmed H, et al. Automatic detection, classification, and segmentation of sagittal MR images for diagnosing prolapsed lumbar intervertebral disc. *Sci Rep* 2025;15(1):593. <https://doi.org/10.1038/s41598-024-84301-7>.
- [19] Nelson J. Roboflow web application. Available from: <https://app.roboflow.com/>. [Accessed 31 March 2025].
- [20] Khanam R, Hussain M. Yolov11: An overview of the key architectural enhancements. *arXiv preprint arXiv:241017725*. 2024. <https://doi.org/10.48550/arXiv.2410.17725>.
- [21] Rao SN. YOLOv11 architecture explained: next-level object detection with enhanced speed and accuracy. Available from: <https://medium.com/@nikhil-rao-20/yolov11-explained-next-level-object-detection-with-enhanced-speed-and-accuracy-2dbe2d376f71>. [Accessed 1 April 2025].
- [22] Yamakawa M, Shiina T, Nishida N, Kudo M. Optimal cropping for input images used in a convolutional neural network for ultrasonic diagnosis of liver tumors. *Jpn J Appl Phys* 2020;59(SK):SKKE09. <https://doi.org/10.35848/1347-4065/ab80dd>.
- [23] Simonyan K, Zisserman A. Very deep convolutional networks for large-scale image recognition. *arXiv preprint arXiv:14091556*. 2014. <https://doi.org/10.48550/arXiv.1409.1556>.
- [24] He K, Zhang X, Ren S, Sun J. Deep residual learning for image recognition. *Proce IEEE confer comput vision pattern recogni* 2016:770–8. <https://doi.org/10.1109/CVPR.2016.90>.
- [25] Xia X, Xu C, Nan B. Inception-v3 for flower classification. 2nd International conference on image, vision and computing (ICIVC). 2017. p. 783–7. <https://doi.org/10.1109/ICIVC.2017.7984661>.
- [26] Huang G, Liu Z, Van Der Maaten L, Weinberger KQ. Densely connected convolutional networks. Proceedings of the IEEE conference on computer vision and pattern recognition. 2017. p. 2261–9. <https://doi.org/10.48550/arXiv.1608.06993>.
- [27] Kim HE, Cosa-Linan A, Santhanam N, Jannesari M, Maros ME, Ganslandt T. Transfer learning for medical image classification: a literature review. *BMC Med Imag* 2022;22(1):69. <https://doi.org/10.1186/s12880-022-00793-7>.
- [28] Abramov M. YOLOv8 vs faster R-CNN: a comparative analysis. Available from: <https://keylabs.ai/blog/yolov8-vs-faster-r-cnn-a-comparative-analysis/>. [Accessed 28 April 2025].
- [29] Sohan M, Sai Ram T, Rami Reddy CV. A review on YOLOv8 and its advancements. *Data Intelligence Cogn Inform* 2024:529–45. https://doi.org/10.1007/978-981-99-7962-2_39.
- [30] Deschenaux J. How robust are pre-trained object detection ML models like YOLO or DETR?. <https://www.lakera.ai/blog/how-robust-are-pre-trained-object-detecti-on-ml-models>. [Accessed 31 March 2025].
- [31] Jahnvi K, Sandeep NS, Deepika R, Battu VSJ, Anitha R, Prakash KB. Detection of COVID-19 using ResNet50, VGG19, MobileNet, and forecasting: using logistic regression, prophet, and SEIRD model. 7th Intern Confer Comput Methodol Commun 2023:23–5. <https://doi.org/10.1109/ICCMC56507.2023.10083564>.
- [32] Latif G, Alghazo J, Khan MA, Ben Braham G, Fawagreh K, Mohammad N. Deep convolutional neural network (CNN) model optimization techniques—review for medical imaging. *AIMS Mathematics* 2024;9(8):20539–71. <https://doi.org/10.3934/math.2024998>.
- [33] Telusdigital. What's the difference between CNN and RNN. Available from: <https://www.telusdigital.com/insights/ai-data/article/difference-between-cnn-and-rnn>. [Accessed 7 April 2025].
- [34] Bansal S. CNN architectures : VGG, ResNet, Inception + TL; 2019. Available from: <https://www.kaggle.com/code/shivamb/cnn-architectures-vgg-resnet-inception-tl>. [Accessed 7 April 2025].
- [35] Sušteršič T, Ranković V, Milovanović V, Kovačević V, Rasulić L, Filipović N. A deep learning model for automatic detection and classification of disc herniation in magnetic resonance images. *IEEE J Biomed Health Inform* 2022;26(12):6036–46. <https://doi.org/10.1109/JBHI.2022.3209585>.
- [36] Salehi E, Khanbare S, Yousefi H, Sharpasand H, Sheymani OS. Deep convolutional neural networks for automated diagnosis of disc herniation on axial MRI. 2019 Sci Meet Electrical-Electron Biomed Eng Comput Sci 2019;1–6. <https://doi.org/10.1109/EBBT.2019.8741895>.

- [37] Lu S, Liu J, Wang X, Zhou Y. Collaborative multi-metadata fusion to improve the classification of lumbar disc herniation. *IEEE Trans Med Imag* 2023;42(12): 3590–601. <https://doi.org/10.1109/tmi.2023.3294248>.
- [38] Ebrahim M, Alsmirat M, Al-Ayyoub M. Advanced disk herniation computer aided diagnosis system. *Sci Rep* 2024;14(1):8071. <https://doi.org/10.1038/s41598-024-58283-5>.
- [39] Tan DS, Nisar H, Yeap KH, Dakulagi V, Amin M. Lumbar intervertebral disc detection and classification with novel deep learning models. *J King Saud Univ Comput Inf Sci* 2024;36(7):102148. <https://doi.org/10.1016/j.jksuci.2024.102148>.